16-Multivariate Analysis of Macrozoobenthos Samples from Two Small Streams

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1 Introduction

The aim of the following exercise is to demonstrate some important multivariate methods by example of a macrozoobenthos data set from two small streams. In some cases, several alternatives are presented, but for a real analysis one does not need to include everything. On the other hand, the methods offer further possibilities that cannot all be presented, see the online help and corresponding books and tutorials.

The data set used originates from a field experiment to investigate the influence of fish predation on the macrozoobenthos species composition. However, during the study period an extreme flood occurred in August and caused major morphological changes to the streams. This motivated the hypothesis that the species community has also changed.

2 Material and Methods

The data set originates from the two streams, "Gauernitzbach" and "Tännichtgrundbach" near Dresden. It contains a table of macrozoobenthos taxa and a description of the sites (t = Tännichtgrund, g = Gauernitz, p = pool, r = riffle) and the time before (v) and after (n)) the flood. The variable names were left in German with Bach (stream), Hochwasser (before and after the flood), Habitat and Site. This may be changed in the future. The data set is a subset from a longer project. To make it accessible as a teaching example, the relatively large number of species of the original data was pragmatically aggregated to a small number of taxa. More about the experiment can be found in (Winkelmann et al., 2008, 2011).

The data set is available from https://tpetzoldt.github.io/datasets/data/gauernitz.csv, together with a file description.

After reading the data in with `read.csv` row names are assigned to the data frame from the `site` column that contains short codes for the observarions, e.g. `GP9` than means Gauernitzbach, pool, September. These rownames are very useful to indicate the observations in the plots.

In a second step, we split the original data frame in two separate tables `bio` with the taxonomic data only and `env` with the environmental factors. For the calculations, it is important that the `bio`-table contains only numbers. The data analysis is carried out with the **vegan** package (Oksanen et al., 2024).

```
library("dplyr")
library("vegan")
dat <- read.csv("gauernitz.csv")

## assign rownames to dataframe
row.names(dat) <- dat$site

## separate into two matrices, bio und env
env <- dat |> select(Habitat, Bach, Hochwasser)
bio <- dat |> select(Mollusca, Diptera, Baetis, Plecoptera, Coleoptera,
             Turbellaria, Heptageniidae, Ephemeroptera, Gammarus,
             Trichoptera, Acari, Nematoda, Oligochaeta)
```

After loading the data, it is always important to have a look at the data structure, for example with `str(bio)`, `str(env)` or in the "Global Environment" pane of RStudio.

3 Data Analysis

3.1 NMDS

We start with an NMDS (nonmetric multidimensional scaling) of the `bio`-data using the Bray-Curtis dissimilarity measure. It is the default of `metaMDS`, but we specify it explicitly to make the selection of the dissimilarity measure clearly visible in the code. Automatic transformation is switched off. This can be changed, depending on the properties of the data, or enabled "manually" for example with `wisconsin(sqrt(bio))`.

The function `metaMDS` then runs the NMDS several times with different starting values to avoid local minima. For difficult data sets, it may be necessary to increase the metaparameters `try` and `trymax`, see helpfile for details.

After that, we should have a look at the stress value and the stressplot.

```
mds <- metaMDS(bio, distance = "bray", autotransform = FALSE)
mds
stressplot(mds)
```

We can then plot the results of the NMDS-ordination.

```
plot(mds, type = "t")
```

In order to show the influence of environmental variables, we can fit vectors or factors to the ordination. In addition to this, we can show the significance of the fitted vectors. For getting reliable p-values, I recommend to increase `permu` to 3999 or 9999.

```
## fit environmental factors and perform a permutations-test
efit <- envfit(mds ~ Hochwasser + Bach + Habitat, env, permu = 999)
efit
```

Now, we can visualize the complete result. Grey dotted zero-lines are added to make interpretation easier.

```
plot(mds, type = "t")
plot(efit, add = TRUE)
abline(h=0, col="grey", lty="dotted")
abline(v=0, col="grey", lty="dotted")
```

3.2 Hierarchical Clustering

The NMDS tries to project the distances as good as possible to a low number of dimensions, e.g. `k=2` that is the default. To see the full picture of distances in multidimensional space, we may consider to apply hierarchical clustering. As agglomeration algorithm, `complete`, `ward.D2` or `ward.D` can be a good choice. To improve understanding it can be a good idea to compare it with other agglomeration schemes, e.g. `single`.

```
hc <- hclust(vegdist(bio), method="ward.D")
plot(hc)
```

It is also possible, to colorize the clusters in the NMDS plot. Let's assume we have 4 clusters, we can first indicate it in the hierarchical cluster tree with `rect.hclust``, then cut the tree with`cutree‘.

```
plot(hc)
rect.hclust(hc, 4)
grp <- cutree(hc, 4)  # assign observations to 4 groups
grp
```

The result is an assignment of the original observations to groups, that can be used to colorize the NMDS plot. It is possible to show the cluster tree directly in the nmds plot or to indicate it otherwise with, for example, `ordispider`, `ordihull` or `ordiellipse`

```
plot(mds, type = "n")
text(mds$points, row.names(bio), col = grp)

## optional: show cluster tree
#ordicluster(mds, hc, col="blue")
```

Exercises: Compare different agglomeration schemes, try different numbers of clusters in `rect.hclust` and `cutree` and add the fitted environmental variables in the final plot.

3.3 Canonical Correspondence Analysis

As an alternative to NMDS, we can also use CCA, that is a "constraied ordination method" and allows a more detailed numerical analysis (e.g. separatation of inertia), but is limited to χ^2-distance, while NMDS allows arbitrary distance measures, including Bray-Curtis.

```
cc <- cca(bio ~ Habitat + Bach + Hochwasser, data = env)
#cc <- cca(bio ~ ., data = env) # same. The . means all from env
cc # print Eigenvalues
plot(cc)
ordihull(cc, env$Habitat, col = "blue")   # or: ordispider, ordiellipse ...
```

3.4 Test of significance

The CCA supports also significance tests and model selection with ANOVA-like permutation tests.

```
## Resampling-ANOVAs of the CCA
anova(cc)
anova(cc, by = "terms") # most useful
anova(cc, by = "axis")

## Model selection to find the optimal model
step(cc)
```

Several other multivariate significance tests exist. The Adonis-Test is in particular popular, because it considers also interaction terms. It does not rely on an NMDS or CCA and works directly with a distance matrix. In order to increase its power, we may optionally consider strata. The following shows some examples.

Exercise: Try different model formulae and decide which one is most appropriate for the data set and the original hypothesis.

```
dist <- vegdist(bio, method = "bray")

adonis2(dist ~ Hochwasser * Habitat * Bach, data = dat, by = "terms")

## Comparison with and without strata
adonis2(dist ~ Hochwasser * Bach, strata = env$Habitat, data = dat, by = "terms")
adonis2(dist ~ Hochwasser * Bach, data = dat, by = "terms")
```

3.5 dbRDA and elimination of covariates

The following examples show further possibilities. Instead of a CCA (that uses χ^2) we can also use a so-called distance-based redundancy analysis (dbRDA), that supports arbitrary distance measures, e.g. Bray-Curtis.

Another option is a partial CCA or partial dbRDA where we can eliminate covariates (`condtion = ...`) that we are not much interested in, so that the ordination focuses on the variables we are interested in. This is the called a partial analysis (pCCA, p-dbRDA). We will then also get three kinds of eigenvalues and eigenvectors (components of the inertia).

```
## ======================================================================
## dbRDA, supports arbitrary dissimilarity measures
## ======================================================================

dbr <- dbrda(dist ~ Habitat + Bach + Hochwasser, data = env, distance = "bray")
dbr
anova(dbr, by="terms", permutations=3999)
#summary(dbr)
plot(dbr)

## ======================================================================
## partial CCA: elimination of covariates
```

```
## ============================================================================
pcc <- cca(bio ~ Hochwasser + Bach + Condition(Habitat), data = env)
pcc
plot(pcc)


## ============================================================================
## partial dbRDA
## ============================================================================
dbrc <- dbrda(bio ~ Hochwasser + Bach + Condition(Habitat),
              data = env, distance = "bray")
dbrc
plot(dbrc)
```

Exercise: Apply a method that eliminates the differences between the streams and investigate whether pools and riffles behave differently.

3.6 Procrustes test

To compare the ordinations, that we get with a different set of environmental variable and conditions, we can use the so-called Procrustes test.

```
dbr  <- dbrda(bio ~ Hochwasser + Bach + Habitat,
              data = env, distance="bray")
pdbr <- dbrda(bio ~ Hochwasser + Bach + Condition(Habitat),
              data = env, distance="bray")

proc <- procrustes(dbr, pdbr)
plot(proc, type = "t")
protest(dbr, pdbr)
```

4 Final recommendations

Multivariate methods are excellent tools for the explorative analysis of larger data sets. However, it is easy to get lost in the variety of methods and plots, and multivariate ordination programs are not always easy to understand. It is therefore a good idea to summarize the main results afterwards with simpler graphics and summary statistics.

Exercise: Create some (e.g. 2-3) bar charts and/or x-y-plots to visualize main results, e.g. abundance of typical taxa or classical diversity indices like Simpson's index and interpret the results.

The following code example shows one of the taxa and the Simpson index for all sites. Now find meaningful aggregations of sites (e.g. stream, pool-riffle, time or cluster) to vizualize main results of the multivariate analysis.

```
barplot(bio$Mollusca, names.arg=dat$Site, horiz=TRUE, las=1)
barplot(diversity(bio, index="simpson"), names.arg=dat$Site, horiz=TRUE, las=1)
```

References

Oksanen, J., Simpson, G. L., Blanchet, F. G., Kindt, R., Legendre, P., Minchin, P. R., O'Hara, R. B., Solymos, P., Stevens, M. H. H., Szoecs, E., Wagner, H., Barbour, M., Bedward, M., Bolker, B., Borcard, D., Carvalho, G., Chirico, M., De Caceres, M., Durand, S., … Weedon, J. (2024). *Vegan: Community ecology package*. https://CRAN.R-project.org/package=vegan

Winkelmann, C., Hellmann, C., Worischka, S., Petzoldt, T., & Benndorf, J. (2011). Fish predation affects the structure of a benthic community. *Freshwater Biology*, *56*(6), 1030–1046. https://doi.org/https://doi.org/10.1111/j.1365-2427.2010.02543.x

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